

STATISTICS — COMPLETE BEGINNER'S GUIDE

Time Series Regression *From Zero to Full Understanding*

Who this guide is for. This document assumes you have never studied statistics before. Every concept is built from the ground up with plain language, visual diagrams, and step-by-step worked examples. Nothing is skipped. If you already have some stats background, the later chapters still add valuable depth and worked examples.

Note on this edition: Four gaps from the original have been filled in, marked with [NEW CONTENT] throughout: (1) the complete ρ -size reference table, (2) the completed four-case summary card, (3) a new section on confidence intervals, and (4) a full set of practice exam questions with answers.

NOTATION GLOSSARY — Read This First

Before anything else, here is every symbol you will encounter, explained in plain English.

Symbol	Name	Plain meaning
Y	Dependent variable	The thing you are trying to predict or explain
X	Independent / explanatory variable	The thing you use to explain Y
t	Time index	Which time period we are in (e.g., $t = 1$ means year 1)
i	Individual index	Which person/firm/country we are looking at
β (beta)	Regression coefficient	How much Y changes when X changes by 1 unit
α (alpha)	Intercept	The value of Y when all X's = 0
ε (epsilon)	Error / residual	The part of Y that our model fails to explain
ρ (rho)	Autocorrelation coefficient	How strongly this period's error relates to last period's error
γ (gamma)	AR coefficient on lagged Y	How much last period's Y feeds into this period's Y
λ (lambda)	Trend slope	How much the mean of Y grows each period
σ^2	Variance	Measure of spread / unpredictability
Δ (delta)	First difference	Change from one period to the next: $\Delta X_t = X_t - X_{t-1}$
Σ (sigma)	Sum	Add up everything that follows
H_0	Null hypothesis	The 'boring' default claim we try to disprove

Symbol	Name	Plain meaning
H_1	Alternative hypothesis	The 'interesting' claim we want to support
n	Sample size	Total number of observations
R^2	R-squared	Proportion of Y's variation explained by the model (0 to 1)
SE	Standard error	How much a coefficient estimate typically varies across samples
CI	Confidence interval	A range of plausible values for the true β (new: see Section 0.4b)
LM	Lagrange Multiplier	A test statistic we compute to check for autocorrelation
χ^2	Chi-squared	A probability distribution used for some hypothesis tests
DW	Durbin-Watson	A quick test statistic for autocorrelation (limited use)

PART 0 — STATISTICS FOUNDATIONS (Start Here)

Everything in this section is background knowledge that the rest of the guide builds on.

0.1 — What Is Statistics? What Is Regression?

Statistics is the science of learning from data. We observe the world, collect numbers, and try to answer questions like: Does more education cause higher wages? Does advertising increase sales? Does training reduce workplace accidents?

The problem is: the world is messy. Wages are affected by education, *and* experience, *and* gender, *and* luck. Statistics lets us separate these influences.

Regression is the central tool. It finds the straight-line relationship between one or more explanatory variables (X's) and an outcome variable (Y), while controlling for noise.

The Simplest Regression (One Variable)

$$Y = \alpha + \beta X + \varepsilon$$

In plain English:

- Y = outcome we want to understand (e.g., hourly wage in euros)
- X = thing we think explains Y (e.g., years of education)
- α = baseline Y when $X = 0$ (e.g., wage with zero education)
- β = slope — how much Y changes for each 1-unit increase in X
- ε = error — everything that moves Y but isn't captured by X

Example: Suppose we find: $\text{Wage} = 5 + 1.2 \times \text{Education} + \varepsilon$

This means: every additional year of education is associated with €1.20/hour more in wages, on average. A person with 10 years of education earns an estimated $5 + 1.2 \times 10 = \text{€}17/\text{hour}$.

The ε (error) is what remains unexplained. If the actual wage is €19/hour and our model predicts €17, the residual is $\varepsilon = 19 - 17 = +2$. Our model is off by €2 for that person.

Multiple Regression (Several Variables)

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

Now we control for several things at once. Each β_j tells us: 'holding all other X's constant, how much does Y change when X_j increases by 1?' This is the **ceteris paribus** (all else equal) interpretation.

0.2 — How Regression Is Estimated: OLS

OLS = Ordinary Least Squares. It is the standard method for calculating the β values.

The idea: choose the values of α and β that make the errors ϵ as small as possible across all observations. Specifically, OLS minimises the **sum of squared errors**:

$$\min \sum \epsilon_i^2 = \sum (Y_i - \alpha - \beta X_i)^2$$

Why square them? Two reasons: 1. Positive and negative errors don't cancel each other out. 2. Large errors are penalised more than small ones (squaring amplifies big mistakes).

OLS gives us **coefficient estimates** $\hat{\beta}$ ('beta hat') — the best-fitting line through the data.

BLUE: When certain conditions hold (see Section 0.4), OLS is **B**est **L**inear **U**nbiased **E**stimator. 'Best' means no other linear method has smaller estimation errors. 'Unbiased' means on average the estimates hit the true value.

0.3 — R^2 : How Well Does the Model Fit?

$$R^2 = 1 - (\text{Sum of Squared Errors}) / (\text{Total Variation in } Y)$$

Range: R^2 is always between 0 and 1.

R^2 value	Meaning
0.0	The model explains nothing — X tells us nothing about Y
0.5	The model explains 50% of Y's variation
1.0	Perfect fit — the model explains everything, zero residual
0.95	Seems great! But in time-series data, this can be a warning sign (spurious regression)

Intuition: Imagine Y is the height of ocean waves. Total variation = how much waves differ from their average height. If your model explains 80% of that variation, $R^2 = 0.80$.

0.4 — Standard Errors, t-statistics, and p-values

Even if OLS gives us $\hat{\beta} = 1.2$, we cannot be certain the true β is 1.2. If we collected a different sample, we'd get a slightly different estimate. **Standard errors (SE)** measure how much $\hat{\beta}$ varies across samples.

The t-statistic

$$t = \hat{\beta} / SE(\hat{\beta})$$

The t-statistic asks: 'Is this estimate far enough from zero to be convincingly non-zero?'

- A **large |t|** (say, $|t| > 2$) means the estimate is large relative to its uncertainty → likely real.
- A **small |t|** (say, $|t| < 1$) means the estimate could easily be zero just by chance → not convincing.

The p-value

The **p-value** is the probability of seeing a t-statistic this extreme (or more) if the true β were actually zero.

p-value	Interpretation
$p < 0.01$	Very strong evidence against H_0 (significant at 1%)

p-value	Interpretation
$p < 0.05$	Strong evidence against H_0 (significant at 5%) — standard threshold
$p < 0.10$	Weak evidence against H_0 (significant at 10%)
$p > 0.10$	No convincing evidence; cannot reject H_0

Analogy: You flip a coin 20 times and get 17 heads. The p-value asks: 'If this were a fair coin (H_0), how likely is getting 17 or more heads?' If that probability is tiny ($p < 0.05$), you conclude the coin is probably not fair — you reject H_0 .

Hypothesis Testing in Regression: For any coefficient β_j : $H_0: \beta_j = 0$ (X_j has no effect on Y) vs. $H_1: \beta_j \neq 0$ (X_j does affect Y). If $p < 0.05$: reject $H_0 \rightarrow X_j$ is statistically significant.

Important phrasing: We never 'accept H_0 '. We either reject it or fail to reject it. Failing to reject just means the data don't give us enough evidence — not that H_0 is definitely true.

[NEW CONTENT]

0.4b — Confidence Intervals [NEW]

A **confidence interval (CI)** gives a range of plausible values for the true population parameter β , based on our sample estimate $\hat{\beta}$. It is the natural complement to the t-test and p-value: while a p-value gives a yes/no decision, the CI tells you *how big* the effect plausibly is.

The formula for a 95% confidence interval:

$$CI = \hat{\beta} \pm 1.96 \times SE(\hat{\beta})$$

This means: we are 95% confident the true β falls somewhere in the range $[\hat{\beta} - 1.96 \times SE, \hat{\beta} + 1.96 \times SE]$.

Confidence level	Multiplier (z^*)	Interpretation
90%	1.645	10% chance the true β is outside this range
95%	1.960	5% chance the true β is outside this range (most common)
99%	2.576	1% chance the true β is outside this range

Worked example: Suppose $\hat{\beta} = 1.20$ and $SE(\hat{\beta}) = 0.35$.

$$95\% CI = 1.20 \pm 1.96 \times 0.35 = 1.20 \pm 0.686 = [0.514, 1.886]$$

Interpretation: We are 95% confident the true effect of one extra year of education on wages is between +€0.51/hr and +€1.89/hr.

Three key rules about confidence intervals:

1. **Link to significance:** If the 95% CI does NOT contain zero, the coefficient is significant at the 5% level — this is exactly equivalent to $p < 0.05$.

2. **Width tells you precision:** A narrow CI means the estimate is precise. A wide CI means you have a lot of uncertainty — perhaps because the sample is small or the data are noisy.

3. **Width increases with SE:** Larger standard errors produce wider intervals. Autocorrelation inflates SE, so it makes CIs wider — another reason to test for and fix autocorrelation.

In time-series models: When autocorrelation is present, the reported SE values are *too small*, which makes CIs artificially narrow. Fixing autocorrelation (via DL or AR re-specification) restores correct SE values and correct CI widths.

0.5 — The Five Classical Assumptions

OLS is only BLUE when these five conditions hold:

Assumption	What it says	Why it matters
A1	$E(\epsilon) = 0$	Errors average to zero — no systematic over/under-prediction
A2	$\text{Var}(\epsilon) = \sigma^2$ (constant)	Homoskedasticity — equal spread of errors everywhere
A3	$\text{cov}(\epsilon_i, \epsilon_j) = 0$	Errors are uncorrelated with each other
A4	X's are non-random, not perfectly correlated	We can separate each X's effect
A5	$\epsilon \sim \text{Normal}$ (optional)	Needed for exact inference in small samples

When you use time-series data, **A3 is routinely violated** (autocorrelation). This is the central problem this guide addresses. A new assumption is also needed: **stationarity**.

0.6 — Visual Intuition: What a Regression Looks Like



Each * is a data point. OLS finds the straight line that passes as close as possible to all the points, minimising the total squared vertical distance from points to the line.

The **residuals** (errors) are the vertical distances from each point to the line. Good regression: residuals look like random noise. Bad regression: residuals show a pattern (e.g., all positive on the left, all negative on the right).

PART 1 — TIME SERIES FOUNDATIONS

CHAPTER TS_A — Cautions

What makes time-series data special and why ordinary regression needs modification

A.1 — Cross-Section vs. Time-Series Data

Definitions

Type	Description	Notation	Order matters?
Cross-section	Many units at one point in time	X_i , subscript i = individual	No — rows are interchangeable
Time-series	One unit observed repeatedly over time	X_t , subscript t = time period	Yes — shuffling destroys information
Panel data	Many units over many time periods	X_{it}	Both dimensions matter

Why Order Matters: In a cross-section of 1,000 workers you could randomly reorder the rows and your regression results would be identical. In a time series of 40 years of GDP data, row 5 (GDP in year 5) must come after row 4 and before row 6. The temporal structure is not a nuisance — it is the entire point.

Cross-section data (each row is a different person):

Person	Wage	Education	Age
Alice	22	16	34
Bob	18	12	28
Carol	35	20	45
...	...	...	...

You could sort Alice, Bob, Carol in any order. Same result.

Time-series data (each row is a different time period):

Year	Belgium GDP	Unemployment	Inflation
2000	253 bn	6.9%	2.7%
2001	259 bn	6.6%	2.4%
2002	263 bn	7.5%	1.6%
2003	268 bn	8.2%	1.5%
...	...	...	...

Row 2003 MUST come after row 2002.

A.2 — Lagged Variables

What a Lag Is: A lag simply refers to the value of a variable at an earlier time period. Think of it as 'looking in the rear-view mirror' — what was the value one period ago? Two periods ago?

Notation	Meaning	Value refers to
X_t	Current value of X	Q3 1987
X_{t-1}	One-period lag	Q2 1987
X_{t-2}	Two-period lag	Q1 1987
X_{t-q}	q -period lag	q quarters before Q3 1987

After creating lag X_{t-1} :

```

t=1: 10 | X_{t-1}: [missing – no period 0]
t=2: 14 | X_{t-1}: 10 <- value from t=1
t=3: 11 | X_{t-1}: 14 <- value from t=2
t=4: 16 | X_{t-1}: 11 <- value from t=3
t=5: 13 | X_{t-1}: 16 <- value from t=4
Notice: first row loses its observation (no 'period 0').

```

Why Lags Are Needed: Economic processes rarely respond instantaneously. Consider safety training: hours are delivered in month t , workers need time to absorb the training, and the reduction in accidents appears in months $t+1$, $t+2$, etc.

Another example — central bank interest rates: When a central bank raises interest rates today, the effect on consumer spending might take 6–12 months to fully materialise.

Watch Out: Lost Observations. Each lag of order q costs you q observations at the start. Lag of order 1 $\rightarrow$ lose 1 observation. Lag of order $q \rightarrow$ lose q observations.

A.3 — Difference Variables

Definition: The first difference of X_t is the change from one period to the next: $\Delta X_t = X_t - X_{t-1}$

Intuition: If GDP was €100 billion last year and €104 billion this year, the first difference is $\Delta \text{GDP} = 104 - 100 = +€4$ billion. It answers: 'How much did GDP change?'

```

GDP level (trending upward – non-stationary):
GDP |
    |
    |
    |
    |
    |
    +----- time
Clearly trending -> not stationary

```

```

ΔGDP (first differences – fluctuates around zero):
ΔGDP |
     | * * *
     | * * * * *
     | ██████████ 0
     | * * *
     |
     +----- time
Fluctuates around a constant level -> stationary

```

Why Logarithms? When you take $\log(\text{price})$ and then difference it: $\Delta \ln(X_t) \approx (X_t - X_{t-1})/X_{t-1}$, you get the **percentage change** — directly comparable across differently-sized quantities.

Why Differences Matter: (1) Economic theory talks about changes. (2) They cure a stochastic trend (unit root). (3) Log-returns are scale-free.

A.4 — The Three Problems with Time-Series Regression

Every time you run a regression on time-series data, you must address three specific issues.

Problem 1 — Data Quality and Sample Size

Situation	Recommendation
Ideal	$n > 100$ observations
Minimum	$n \approx 35$; at least 10 obs per X-variable in the model
Yearly data	May need 35+ years; consider monthly or quarterly if available
Monetary variables	Use real (deflated) values — inflation creates artificial trends

Why inflation matters: If you regress nominal wages on nominal GDP, both series may trend upward simply because of inflation, not because of any real relationship. Always use inflation-adjusted (real) values for monetary variables.

Problem 2 — Autocorrelation

Classical Assumption A3 states that errors are uncorrelated: $\text{cov}(\epsilon_t, \epsilon_s) = 0$. In time-series data this is routinely violated. Shocks have persistence — if something unexpected happens today, its effect persists into next period, gradually fading.

Problem 3 — Spurious Regression / Non-Stationarity

Imagine regressing Belgian ice cream sales on annual sunspot activity. Both series happen to trend upward over a 30-year window. You would find a hugely significant regression with $R^2 = 0.95$ — even though they have absolutely nothing to do with each other.

The solution is to ensure all variables are stationary before fitting the regression.

CHAPTER TS_B — Autocorrelation

Errors that remember the past: detection with the LMSC test

B.1 — Which Classical Assumption is Violated?

Autocorrelation violates **A3**. When errors are correlated over time, $\text{cov}(\varepsilon_t, \varepsilon_{t-1}) \neq 0$.

Why A3 Fails So Commonly in Time Series: Think of the residual ε_t as 'everything that moves Y but isn't in my model.' In time-series settings, omitted variables themselves tend to persist over time. Today's omitted shock tends to carry over into tomorrow.

Intuitive example: You model monthly consumer spending using only income. But consumer confidence, which you haven't measured, also drives spending — and consumer confidence is persistent. The residuals will therefore form waves: positive for a stretch (when confidence is high), then negative for a stretch (when confidence is low). That is positive autocorrelation.

B.2 — First-Order Autocorrelation

The AR(1) Error Structure: $\varepsilon_t = \rho \varepsilon_{t-1} + u_t$

where: ρ (rho) is the first-order autocorrelation coefficient (between -1 and 1); u_t is classical white-noise error (uncorrelated, mean zero, constant variance).

White noise means pure randomness — like rolling a fair die each period. No memory, no pattern, no trend.

No autocorrelation ($\rho = 0$) — ideal:

```
Residuals
| * * * *
| * * * *
0 | ████████████████████████████████████████████
| * * * * *
| * *
Random scatter. A3 satisfied. ✓
```

Positive autocorrelation ($\rho > 0$) — 'waves':

```
Residuals
| * * * *
| * * * *
0 | ████████████████████████████████████████████
| * * * *
| * * *
Long runs of + then long runs of -. A3 violated. ✗
```

Negative autocorrelation ($\rho < 0$) — 'zigzag':

```
Residuals
| * * * *
0 | ████████████████████████████████████████████
| * * * *
Alternates above/below every period. Rare. A3 violated. ✗
```

[NEW CONTENT]

Understanding the Size of ρ — Complete Reference Table [NEW]

The magnitude of ρ tells you how severe the autocorrelation is. This table was incomplete in the original; here is the full reference:

Value of ρ	Autocorrelation strength	Effect on residuals	Practical consequence
$\rho = 0$	None	Pure white noise — ideal	A3 satisfied; OLS is BLUE
$0 < \rho \leq 0.3$	Weak positive	Short runs of same sign	Minor inflation of t-stats; often tolerable
$0.3 < \rho \leq 0.6$	Moderate positive	Noticeable wave pattern	Standard errors meaningfully wrong; fix required
$0.6 < \rho < 1$	Strong positive	Long persistent waves	Severe bias in SEs; t-tests very misleading
$\rho \rightarrow 1$	Near unit root	Series barely returns to mean	Borders on non-stationarity; re-examine series
$-0.3 \leq \rho < 0$	Weak negative	Slight alternation	Rare in economics; minor effect
$\rho < -0.3$	Strong negative	Rapid alternation + – + –	Often caused by over-differencing

Key boundary: We require $-1 < \rho < 1$ for the error process to be stationary and well-behaved.

Worked analogy for $\rho = 0.7$: If this quarter's residual is +10, next quarter's expected residual is +7 ($= 0.7 \times 10$). Then +4.9, then +3.4, then +2.4 — slowly decaying toward zero. A shock takes many periods to fade.

B.3 — Consequences of Autocorrelation

Property	With autocorrelation
OLS estimators β	Still unbiased — on average they hit the true β
Efficiency	No longer best (not BLUE) — a better estimator exists
Standard errors	Wrong — they use the wrong formula (assumes A3)
t-tests and F-tests	Invalid — built on wrong standard errors
Confidence intervals	Wrong width — too narrow if $\rho > 0$ (falsely precise)
R^2	Unaffected but potentially misleading

The key danger is **false significance**: positive autocorrelation inflates t-statistics, making coefficients appear more significant than they really are.

Analogy for wrong standard errors: Imagine asking 10 people in one household whether they like pizza, and claiming those 10 opinions represent 10 independent data points. They don't — family members influence each other. Autocorrelation does the same thing: consecutive observations are not truly independent, so your effective sample size is smaller than it appears.

Exception: AR(1) Models. When the regression contains Y_{t-1} as a regressor AND there is autocorrelation, the coefficient estimates themselves become **biased**. **Practical rule:** If you detect autocorrelation in an AR(1) model, re-specify (switch to DL(q)).

B.4 — Detection: The LMSC Test

Step 0 — Graphical Inspection First. Always plot the residuals against time before running any test.

Hypotheses: H_0 : $\rho = 0$ (no autocorrelation — A3 satisfied) vs. H_1 : $\rho \neq 0$ (autocorrelation present — A3 violated)

Step 1: Fit the original regression model. Save the residuals e_t .

Step 2: Build the auxiliary regression. Regress residuals on all original predictors $X_{\{1t\}}, \dots, X_{\{kt\}}$ plus the lagged residual $e_{\{t-1\}}$:

$$e_t = \alpha_0 + \alpha_1 X_{\{1t\}} + \dots + \alpha_k X_{\{kt\}} + \alpha_{k+1} e_{\{t-1\}} + u_t$$

Step 3: From the auxiliary regression, compute: $LM = n \times R^2$

Step 4: Under H_0 , $LM \sim \chi^2(1)$. Compare LM to **3.841** (5% critical value for $\chi^2(1)$).

Decision: If $LM > 3.841$ (or $p < 0.05$): Reject $H_0 \rightarrow$ autocorrelation is present. If $LM \leq 3.841$ (or $p \geq 0.05$): Do not reject $H_0 \rightarrow A3$ plausibly satisfied.

Worked Example — Phillips Curve

Original model: $Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t$ — Y = inflation rate (%), X = change in unemployment rate (pp) — $n = 90$ quarterly observations

Item	Value
n for auxiliary regression	$90 - 1 = 89$
R^2 of auxiliary regression	0.310211
Coefficient on $e_{\{t-1\}}$	0.558 ($t = 6.219$, $p < .001$)
LM statistic	$89 \times 0.310211 = 27.609$
Critical value (5%)	3.841
Conclusion	Reject H_0 — strong autocorrelation present

Durbin–Watson as an Alternative

DW value	Interpretation
Close to 2	No autocorrelation
Close to 0	Strong positive autocorrelation (ρ near +1)
Close to 4	Strong negative autocorrelation (ρ near -1)
1.5 to 2.5	Generally acceptable (rough rule)

Critical limitation: Durbin–Watson is **invalid** whenever the regression contains a lagged dependent variable $Y_{\{t-1\}}$. Always use LMSC as your primary test.

Relationship: $DW \approx 2(1 - \rho)$. So $DW = 2$ when $\rho = 0$, $DW = 0$ when $\rho = 1$.

B.5 — Solutions for Autocorrelation

Re-specification	What you add	Model type
Add lagged X's	$X_{\{t-1\}}, X_{\{t-2\}}, \dots, X_{\{t-q\}}$	DL(q) — Distributed Lag
Add lagged Y	$Y_{\{t-1\}}$	AR(1) — Autoregressive

Key principle: bias is worse than inefficiency. DL with autocorrelation $\rightarrow$ wrong standard errors (bad, but coefficients are still correct on average). AR with autocorrelation $\rightarrow$ wrong coefficient estimates (fundamentally misleading).

CHAPTER TS_C — Stationarity

Spurious regression, the generalized AR(1), and the Dickey-Fuller test

C.1 — Spurious Regression

The Problem: Consider: regress Belgian GDP per capita (1950–2004) on US population density over the same period. You find $R^2 \approx 0.96$ (extremely high) and slope coefficient significant at $p < 0.001$. Yet Belgian GDP and US population density have no meaningful economic relationship. This is a **spurious regression**.

Famous real-world example: The number of Nicolas Cage films released per year is significantly correlated with drowning deaths in US swimming pools between 1999 and 2009. A regression gives $R^2 \approx 0.67$ and $p < 0.05$. Completely spurious.

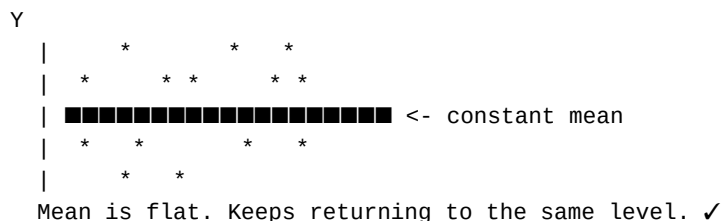
The Root Cause: Non-Stationarity. Both variables are non-stationary — their means drift over time. Standard regression theory requires stationarity.

C.2 — Definition of Stationarity

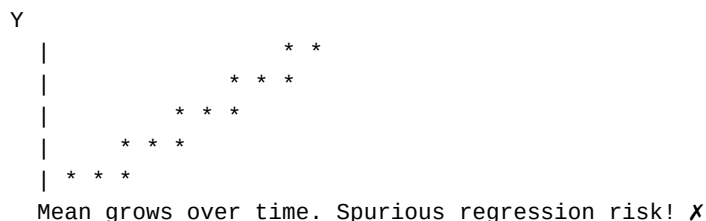
Formal Definition (Weak/Covariance Stationarity): A time series Y_t is weakly stationary if, for every t :

1. $E(Y_t) = \mu$ — The mean is constant.
2. $\text{Var}(Y_t) = \sigma^2$ — The variance is constant.
3. $\text{Cov}(Y_t, Y_{t-s}) = \gamma_s$ — The covariance depends only on the gap s , not on when.

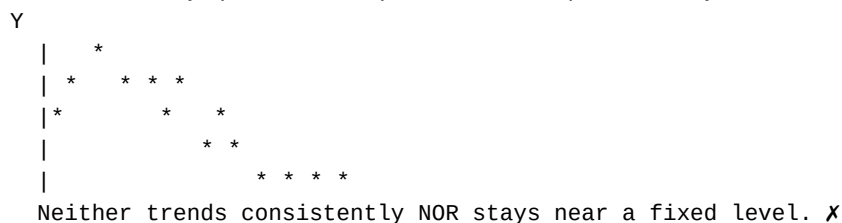
Stationary series – fluctuates around a constant level:



Non-stationary (trending) series – drifts upward:



Non-stationary (random walk) – wanders unpredictably:



C.3 — The Generalized AR(1) Model

$$Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$$

The three parameters: α = constant; λ = trend slope; ρ = autoregressive coefficient. The critical question: $|\rho| < 1$ (stationary) or $|\rho| = 1$ (random walk)?

Case 1 — Basic AR(1): $|\rho| < 1$, $\alpha = 0$, $\lambda = 0$

Model: $Y_t = \rho Y_{t-1} + \varepsilon_t$. Each period, Y reverts toward zero. Properties: Mean = 0 (constant ✓); Var = $\sigma^2/(1-\rho^2)$ (constant ✓); stationary.

Example: $Y_t = 0.7 Y_{t-1} + \varepsilon_t$. A shock of +10: $t+1$: +7.0, $t+2$: +4.9, $t+3$: +3.4, ... $\rightarrow 0$. Shocks fade.

Case 2 — AR(1) with constant: $|\rho| < 1$, $\alpha \neq 0$, $\lambda = 0$

Model: $Y_t = \alpha + \rho Y_{t-1} + \varepsilon_t$. Long-run mean $\mu = \alpha/(1-\rho)$. **Stationary** around non-zero mean.

Example: $Y_t = 5 + 0.6 Y_{t-1} + \varepsilon_t$. Long-run mean = $5/(1-0.6) = 12.5$.

Case 3 — AR(1) with deterministic trend: $|\rho| < 1$, $\alpha \neq 0$, $\lambda \neq 0$

Model: $Y_t = \alpha + \lambda t + \rho Y_{t-1} + \varepsilon_t$. The λt term makes the mean grow linearly. **Not stationary in levels**, but deviation from the trend line is stationary. **Fix:** Add t as a regressor.

Case 4 — Random Walk: $|\rho| = 1$

When $\rho = 1$, $Y_t = \alpha + Y_{t-1} + \varepsilon_t$. Each shock is **permanent**. By repeated substitution: $Y_t = Y_0 + \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t$.

Variance = $t\sigma^2 \rightarrow$ **grows without bound**. **Fix:** Take first differences.

[NEW CONTENT]

Four-Case Summary Card — Completed [NEW]

The original summary card had missing entries. Here is the complete version:

Case	ρ	λ	Stationary?	Correct Fix
Basic AR(1)	$ \rho < 1$	$= 0$	✓ Yes	No fix needed — use in levels
AR(1) + constant	$ \rho < 1$	$= 0$	✓ Yes	No fix needed — use in levels
AR(1) + deterministic trend	$ \rho < 1$	$\neq 0$	✗ No (in levels) ✓ About trend line	Add t as regressor (Option: detrend Y first)
Random walk (pure, $\rho = 1$, $\alpha = 0$)	$= 1$	$= 0$	✗ No (variance grows)	Take first differences ΔY_t
Random walk with drift ($\rho = 1$, $\alpha \neq 0$)	$= 1$	$= 0$	✗ No (mean drifts)	Take first differences ΔY_t

Memory rule for the fix:

- Series wanders with permanently accumulating shocks ($\rho = 1$) $\rightarrow$ **First difference**
- Series trends predictably but returns to trend after shocks ($|\rho| < 1$, $\lambda \neq 0$) $\rightarrow$ **Add t as regressor**
- Series is already mean-reverting ($|\rho| < 1$, $\lambda = 0$) $\rightarrow$ **No fix needed**

C.4 — Deterministic vs. Stochastic Trends

This distinction is crucial because the two types of non-stationarity require **different fixes**. Applying the wrong fix can make things worse.

Feature	Deterministic Trend	Stochastic Trend (Random Walk)
Mathematical source	λt term in the model	$\rho = 1$ (unit root)

Feature	Deterministic Trend	Stochastic Trend (Random Walk)
Mean	Changes predictably (grows by λ per period)	May drift, but unpredictably
Variance	Constant	Grows with time
Shocks	Temporary — series returns to trend	Permanent — series never recovers
Example	GDP trend growth of 2% per year	S&P; 500 stock price index
Correct fix	Detrend or add t as regressor	First difference
Wrong fix	First-differencing $\rightarrow$ negative autocorrelation	Detrending doesn't remove stochastic trend

C.5 — Solutions for Non-Stationarity

Fix A — Deterministic Trend ($|\rho| < 1, \lambda \neq 0$)

Option 2 — Model the trend explicitly (preferred): Include t as an additional regressor in your main model:

$$Y_t = \beta_0 + \beta_1 X_t + \lambda t + \varepsilon_t$$

The coefficient on t absorbs the linear trend; the coefficient on X_t now measures the relationship *after controlling for the common time trend*.

Fix B — Stochastic Trend / Unit Root ($|\rho| = 1$)

Take first differences: $\Delta Y_t = Y_t - Y_{t-1}$. This is the **only valid fix** for a random walk.

Why it works: If $Y_t = Y_{t-1} + \varepsilon_t$, then $\Delta Y_t = \varepsilon_t \leftarrow$ white noise, stationary. ✓

Important caveat — Cointegration: There is one exception. If two non-stationary series are cointegrated — genuinely tied together by an economic relationship — you should *NOT* difference them. Instead, run the regression in levels. Differencing cointegrated variables throws away the most economically meaningful information.

C.6 — The Dickey-Fuller Test

Purpose: The Dickey-Fuller test is a formal unit root test: it decides whether $\rho = 1$ (non-stationary) or $|\rho| < 1$ (stationary).

Hypotheses:

- H_0 : $\rho = 1$ — the series has a unit root, is NOT stationary
- H_1 : $|\rho| < 1$ — the series IS stationary (possibly about a deterministic trend)

Note the asymmetry: H_0 is the dangerous case (non-stationarity). Failing to reject H_0 means we cannot prove stationarity — assume the series needs differencing.

The Reparameterisation: Subtract Y_{t-1} from both sides to get: $\Delta Y_t = a + \lambda t + \alpha Y_{t-1} + \varepsilon_t$ where $\alpha = \rho - 1$. Now H_0 : $\alpha = 0$ and H_1 : $\alpha < 0$.

Why the Normal t-Table Cannot Be Used Here: Under H_0 , Y_t is a random walk — not stationary. The usual asymptotic results do not apply. The actual distribution is shifted left. If you used ± 1.96 , you would almost never reject H_0 for truly stationary series. Dickey and Fuller derived special critical values through simulation:

DF Version	1% critical value	5% critical value	10% critical value
Version 1 (no trend)	-3.43	-2.86	-2.57
Version 2 (with trend)	-3.96	-3.41	-3.13

Decision rule: Reject H_0 (conclude stationarity) if and only if the t-statistic on Y_{t-1} is **more negative** than the critical value.

Version	Regression	When to use
Version 1 — no trend	$\Delta Y_t = \alpha + \alpha Y_{t-1} + \varepsilon_t$	No visible trend in time plot
Version 2 — with trend	$\Delta Y_t = \alpha + \lambda t + \alpha Y_{t-1} + \varepsilon_t$	Clear trend in time plot, or when unsure

Version 2 is the safer default.

DF result	Action
Reject H_0 (stationary)	Use the series in levels in your model
Fail to reject H_0 (unit root)	Take first differences, then re-run DF on ΔY_t
DF on ΔY_t : Reject H_0	ΔY_t is stationary; use first differences in the model
DF on ΔY_t : Still fail to reject	Rare. Consider second differences

CHAPTER TS_D — Dynamic Models

DL(q), AR(1), total multipliers, and the time-series exam workflow

D.1 — Distributed Lag Model DL(q)

A **static model** assumes the entire effect of X on Y is instantaneous: $Y_t = \alpha + \beta X_t + \varepsilon_t$. But most economic effects have delayed responses.

The **Distributed Lag model of order q** captures these delays:

$$Y_t = \alpha + \beta X_t + \beta X_{t-1} + \beta X_{t-2} + \dots + \beta_q X_{t-q} + \varepsilon_t$$

Visual — how DL(2) distributes the effect:

Training hours in month t

```
|
| --- β ---> Effect on accidents in month t    (immediate)
| --- β ---> Effect on accidents in month t+1 (1-period lag)
| --- β ---> Effect on accidents in month t+2 (2-period lag)
```

Total effect of one training hour = $\beta + \beta + \beta$ (total multiplier)

The Total Multiplier: Total multiplier = $\beta + \beta + \dots + \beta_q = \sum \beta_k$

Interpretation: The long-run effect on Y of a **permanent** 1-unit increase in X. 'If X goes up by 1 unit and stays there forever, by how much will Y eventually change?'

D.2 — Choosing the Optimal Lag Length q

Including too many lags creates: (1) lost observations, (2) multicollinearity, (3) overfitting.

The Top-Down Procedure: Start from $q_{\max}$ and progressively drop the highest non-significant lag:

1. Fit DL($q_{\max}$). Check the t-test for $\beta_{q_{\max}}$.
2. If $\beta_{q_{\max}}$ is not significant: drop lag $q_{\max}$, refit DL($q_{\max} - 1$).
3. Repeat until the highest remaining lag is significant.
4. That is your optimal lag length q.

Important: Do not drop intermediate insignificant lags. If β_2 is significant but β_1 is not, keep both — you cannot have lag 2 without lag 1.

Worked Example — Safety Training

Step	Model fitted	Result	Action
1	DL(4)	$\beta_4: t = 0.83, p = 0.41$	Not significant → Drop lag 4
2	DL(3)	$\beta_3: t = 1.21, p = 0.23$	Not significant → Drop lag 3
3	DL(2)	$\beta_2: t = 1.56, p = 0.12$	Not significant → Drop lag 2
4	DL(1)	$\beta_1: t = -2.87, p = 0.005$	Significant → Stop. Optimal q = 1

D.3 — Autoregressive Model AR(1)

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

By including Y_{t-1} , we capture all persistence effects in a single coefficient γ .

Coefficient	Interpretation
α	Intercept
β	Immediate effect: a 1-unit rise in X_t causes Y_t to rise by β , holding Y_{t-1} constant
γ	Persistence coefficient: how much of last period's Y carries forward. Requires $ \gamma < 1$
$\beta/(1-\gamma)$	Total multiplier: long-run effect of a permanent 1-unit rise in X

Proof of the total multiplier formula: In long-run equilibrium, $Y_t = Y_{t-1} = Y^*$. Then: $Y^* = \alpha + \beta X^* + \gamma Y^*$
 $\rightarrow Y^*(1-\gamma) = \alpha + \beta X^* \rightarrow Y^* = \alpha/(1-\gamma) + [\beta/(1-\gamma)]X^*$. So a 1-unit permanent increase in X^* raises long-run Y^* by $\beta/(1-\gamma)$.

Dynamic Path After a Temporary Shock

Period	Additional effect on Y (above baseline)
t	β
$t+1$	$\gamma \cdot \beta$
$t+2$	$\gamma^2 \cdot \beta$
$t+3$	$\gamma^3 \cdot \beta$
$t+k$	$\gamma^k \cdot \beta$

Worked Example — Education Spending and GDP Growth

Fitted AR(1): $\Delta_t = 1.01 + 0.009 X_t + 0.627 Y_{t-1}$

Component	Value	Meaning
α	1.01	Baseline growth when $X = 0$ and $Y_{t-1} = 0$
β	0.009	A \$1 increase raises GDP growth by 0.009 pp immediately
γ	0.627	62.7% of last year's GDP growth persists into this year
Total multiplier	$0.009/(1-0.627) = 0.024$	Permanent \$1/child increase raises long-run GDP growth by 0.024 pp

D.4 — Link Between AR(1) and DL(∞)

AR(1) is equivalent to a DL(∞) with geometrically declining coefficients: $\beta, \beta\gamma, \beta\gamma^2, \beta\gamma^3, \dots$

Total multiplier = $\sum \beta\gamma^k = \beta \times 1/(1-\gamma) = \beta/(1-\gamma)$

Why does $\sum \gamma^k = 1/(1-\gamma)$? Geometric series: $S = 1 + \gamma + \gamma^2 + \gamma^3 + \dots \rightarrow S - \gamma S = 1 \rightarrow S(1-\gamma) = 1 \rightarrow S = 1/(1-\gamma)$. Only converges when $|\gamma| < 1$.

Feature	DL(q)	AR(1)
Lag structure	$q+1$ free coefficients $\beta_0, \beta_1, \dots, \beta_q$	Geometric decay — only β and γ
Parameters needed	$q+1$	2
Multicollinearity	High for large q	Low (only one lag of Y needed)
Bias when A3 violated	Unbiased (only inefficient)	Biased
Lag structure flexibility	Completely flexible	Constrained to geometric decay

Feature	DL(q)	AR(1)
When to prefer	Autocorrelation suspected	Parsimony matters and A3 is satisfied

D.5 — The Three-Step Exam Workflow

Step 1 — Stationarity Check (Chapter TS_C)

1. Plot the series against time.
2. Choose DF version (no visible trend → V1; clear trend or unsure → V2).
3. Run the DF regression, read the t-statistic for Y_{t-1} .
4. Compare to -2.86 (V1 at 5%) or -3.41 (V2 at 5%).
5. Reject H_0 → series is stationary → use in levels.
6. Fail to reject H_0 → unit root → take ΔY_t , re-test.

Step 2 — Fit the Model (Chapter TS_D)

DL(q): Start with $q = q_{\max}$. Drop highest non-significant lag. Repeat. Report all coefficients and total multiplier = $\sum \beta_k$.

AR(1): Fit $Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$ directly. Report β , γ , and total multiplier = $\beta/(1-\gamma)$.

Step 3 — Assumption Check with LMSC (Chapter TS_B)

1. State hypotheses: $H_0: \rho = 0$ vs. $H_1: \rho \neq 0$.
2. Write auxiliary regression: $e_t = \alpha_0 + \alpha_1 X_{1t} + \dots + \alpha_{k+1} e_{t-1} + u_t$.
3. Compute $LM = n_{\text{aux}} \times R^2_{\text{aux}} \sim \chi^2(1)$ under H_0 .
4. Compare to 3.841.
5. Conclude with a full sentence.

FORMULA SHEET — The Five Equations You Must Know

1. DL(q) Model

$$Y_t = \alpha + \beta_1 X_t + \beta_2 X_{t-1} + \dots + \beta_q X_{t-q} + \varepsilon_t$$

$$\text{Total multiplier} = \beta_1 + \beta_2 + \dots + \beta_q = \sum \beta_k$$

2. AR(1) Model

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

$$\text{Total multiplier} = \beta/(1-\gamma) \text{ (requires } |\gamma| < 1)$$

3. LMSC Test

$$\text{Auxiliary regression: } e_t = \alpha_0 + \alpha_1 X_{1t} + \dots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t$$

$$LM = n \cdot R^2 \sim \chi^2(1) \text{ under } H_0$$

$$H_0: \rho = 0 \mid H_1: \rho \neq 0 \mid \text{Critical value at 5\%: } \chi^2(1) = 3.841$$

4. Dickey-Fuller Test

$$\text{Version 1 (no trend): } \Delta Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \varepsilon_t$$

$$\text{Version 2 (with trend): } \Delta Y_t = \alpha_0 + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t$$

$$H_0: \alpha_1 = 0 \text{ (unit root)} \mid H_1: \alpha_1 < 0 \text{ (stationary)}$$

Reject H_0 if t-stat is more negative than critical value (−2.86 or −3.41).

[NEW CONTENT]

5. Confidence Interval [NEW]

$$95\% \text{ CI for } \beta: \beta \pm 1.96 \times SE(\beta)$$

$$\text{General CI: } \beta \pm z^* \times SE(\beta) \text{ where } z^* = 1.645 (90\%), 1.960 (95\%), 2.576 (99\%)$$

$$\text{If } 0 \text{ is outside the 95\% CI } \leftrightarrow \text{coefficient is significant at 5\% } \leftrightarrow p < 0.05$$

EXAM-DAY CHECKLIST

Step 1 — Stationarity (TS_C)

- Plot every variable against time
- Choose DF version (V1: no trend / V2: trend or unsure)
- Run DF on each variable, read t-stat on Y_{t-1}
- Compare to adapted critical values (NOT standard t-table)
- Unit root present? → take ΔY_t , re-test on differences

Step 2 — Fit the Model (TS_D)

- Model type given in question (static / DL(q) / AR(1))
- For DL: start at $q_{\max}$, drop top lag while not significant
- Report all coefficients with standard errors and t-stats
- Calculate and report total multiplier ($\sum \beta_k$ or $\beta/(1-\gamma)$)
- Interpret: distinguish temporary vs. maintained; levels vs. log-returns

Step 3 — Assumption Check (TS_B)

- State H_0 and H_A explicitly
 - Write the auxiliary regression (all original X's + lagged residual)
 - Compute $LM = n_{\text{aux}} \times R^2_{\text{aux}}$
 - Compare to $\chi^2(1) = 3.841$ (or use p-value)
 - Write a full conclusion sentence
-

COMMON EXAM PITFALLS

1. "Reject H_0 means non-stationary" — WRONG

- For the Dickey-Fuller test: Reject H_0 means the series IS stationary (H_0 is the unit root).
- For the LMSC test: Reject H_0 means autocorrelation IS present (H_0 is no autocorrelation).
- These are opposite! Know which test you are running before concluding.

2. "Take differences whenever the series trends" — BE CAREFUL

- Only take differences if the DF test confirms a stochastic trend (unit root, $|\rho| = 1$).
- If the trend is deterministic ($|\rho| < 1$, $\lambda \neq 0$), add t as a regressor instead.
- Over-differencing creates artificial negative autocorrelation and loses information.

3. "Use Durbin-Watson for all models"

- DW is invalid whenever Y_{t-1} appears as a regressor (AR(1) models).
- Always use LMSC — it works for static, DL(q), and AR(1) models.

4. "Total multiplier = β only"

- For DL(q): total multiplier = $\beta_0 + \beta_1 + \dots + \beta_q$ (sum of ALL lag coefficients).
- For AR(1): total multiplier = $\beta/(1-\gamma)$, not just β .
- β_0 alone is the immediate effect only.

5. "Unit changes when X is log-differenced"

- If you differenced $\log(X)$, a 1-unit rise in $\Delta \ln(X)$ means a 1 percentage point rise in the growth rate — not a 1-unit rise in the level.
- Interpret in terms of percentage changes.

6. "Skip the autocorrelation test to save time"

- The LMSC test is worth dedicated exam marks.
- Always report all five elements: H_0/H_1 , the auxiliary regression, $LM = n \times R^2$, comparison to 3.841, and conclusion.

7. "Use the regular t-table for the Dickey-Fuller t-statistic"

- The DF t-statistic does NOT follow a t-distribution under H_0 .
- Standard critical values (± 1.96 , ± 2.58) do not apply.
- Always use the adapted DF critical values: -2.86 (V1, 5%), -3.41 (V2, 5%).

8. "A 95% CI that is very narrow always means a reliable result" [NEW]

- If autocorrelation is present, standard errors are too small $\rightarrow$ CIs are artificially narrow.
- A narrow CI is only trustworthy after you have confirmed A3 is satisfied (LMSC test).
- Always complete Step 3 before placing confidence in your interval estimates.

[NEW CONTENT]

PRACTICE EXAM QUESTIONS [NEW]

Work through these questions yourself before checking the answers. They are structured to match common exam formats for time-series regression.

Question 1 — Identifying Autocorrelation

A researcher fits the model: $Y_t = \alpha + \beta X_t + \varepsilon_t$ using 80 quarterly observations. The Durbin-Watson statistic is 0.72. She then runs the LMSC auxiliary regression and obtains $R^2 = 0.38$ with $n_{\text{aux}} = 79$.

(a) What do you conclude from the DW statistic? Why is this not sufficient?

(b) Compute the LM statistic. What do you conclude at the 5% significance level?

(c) What is the recommended remedy? Why is switching to an AR(1) model risky here?

Answer:

(a) $DW = 0.72$ is close to 0, suggesting strong positive autocorrelation ($\rho \approx 1 - 0.72/2 \approx 0.64$). However, DW alone is not sufficient as a formal test and is inappropriate if any lagged Y is included as a regressor. The LMSC test is always preferred.

(b) $LM = 79 \times 0.38 = 30.02$. Since $30.02 > 3.841$ (the 5% critical value for $\chi^2(1)$), we reject H_0 ($\rho = 0$). Autocorrelation is strongly present; A3 is violated.

(c) Recommended remedy: add lagged X 's — switch to a $DL(q)$ model. Switching to AR(1) is risky because if autocorrelation persists in an AR(1) model, the coefficient estimates become biased (Y_{t-1} becomes correlated with ε_t). DL with autocorrelation is only inefficient, not biased — a much lesser problem.

Question 2 — Stationarity and the Dickey-Fuller Test

You have annual data on a country's inflation rate (1960–2019, $n = 60$ observations). The time plot shows the series fluctuating around a roughly constant level with no clear trend. You run the Dickey-Fuller test (Version 1) and obtain a t-statistic of -3.12 on the coefficient of Y_{t-1} .

(a) State the null and alternative hypotheses of the Dickey-Fuller test.

(b) Is the series stationary at the 5% significance level? At the 1% level?

(c) Why can you NOT use the standard t-table critical value of -1.96 ?

(d) Should you use Version 1 or Version 2? Is your choice here appropriate?

Answer:

(a) $H_0: \alpha = 0$ (series has a unit root — not stationary) vs. $H_1: \alpha < 0$ (series is stationary). Note: this is a one-sided test.

(b) At 5%: critical value for V1 is -2.86 . Since $-3.12 < -2.86$, we reject $H_0 \rightarrow$ series is stationary at 5%. At 1%: critical value for V1 is -3.43 . Since $-3.12 > -3.43$, we fail to reject $H_0 \rightarrow$ not significant at 1%. Conclusion: stationary at 5% but not at 1%.

(c) Under H_0 , Y_t is a random walk — non-stationary. The standard asymptotic theory that justifies the t-distribution assumes stationarity, which fails here. The t-distribution of the DF statistic under H_0 is shifted to the left relative to the standard distribution. Using -1.96 would almost never reject H_0 , even for genuinely stationary series.

(d) Version 1 (no trend) is appropriate here because the time plot showed no clear trend. If there were a visible trend, Version 2 would be required. Using Version 1 on a trending series is a misspecification; using Version 2 on a non-trending series merely loses one degree of freedom — so Version 2 is the safer default when unsure.

Question 3 — Distributed Lag Model and Total Multiplier

Monthly data on advertising spend (X, €000s) and product sales (Y, units) over $n = 120$ months. You fit DL models with $q_{\max} = 3$. Results:

Model	Highest lag coefficient	t-statistic	p-value
DL(3)	$\beta_{\text{■}} = -0.21$	-0.88	0.381
DL(2)	$\beta_{\text{■}} = 0.19$	1.05	0.296
DL(1)	$\beta_{\text{■}} = 0.34$	2.41	0.017

The final DL(1) model gives: $\text{■}_t = 120 + 0.85 X_t + 0.34 X_{t-1}$

(a) Determine the optimal lag length q . Justify your answer.

(b) Interpret the coefficient $\beta_{\text{■}} = 0.85$.

(c) Calculate the total multiplier and interpret it.

(d) Advertising spend is permanently increased by €10,000 starting in month t . What is the total long-run increase in expected monthly sales?

Answer:

(a) Optimal $q = 1$. Starting from $q_{\max} = 3$: $\beta_{\text{■}}$ is not significant ($p = 0.381 > 0.05$) → drop lag 3. $\beta_{\text{■}}$ is not significant ($p = 0.296 > 0.05$) → drop lag 2. $\beta_{\text{■}}$ is significant ($p = 0.017 < 0.05$) → stop. Optimal $q = 1$.

(b) $\beta_{\text{■}} = 0.85$: a €1,000 increase in advertising spend in month t is associated with an immediate increase of 0.85 units in sales in month t , holding all other lagged advertising values constant (ceteris paribus).

(c) Total multiplier = $\beta_{\text{■}} + \beta_{\text{■}} = 0.85 + 0.34 = 1.19$. Interpretation: a permanent €1,000 increase in advertising spend is associated with a long-run increase of 1.19 units in monthly sales.

(d) A permanent increase of €10,000 ($= 10 \times \text{€1,000}$) → total long-run increase = $10 \times 1.19 = 11.9$ units per month.

Question 4 — AR(1) Model

Annual data on a firm's accident rate (Y, incidents per 1,000 workers) as a function of safety training hours (X) — $n = 45$ years. The LMSC test on a static model gives $LM = 8.3$. You switch to an AR(1) model and obtain:

$$\text{■}_t = 12.4 - 0.031 X_t + 0.52 Y_{t-1}$$

Standard errors: $SE(\beta_{\text{■}}) = 0.010$, $SE(\gamma_{\text{■}}) = 0.11$. LMSC on AR(1) residuals: $R^2 = 0.021$, $n_{\text{aux}} = 44$.

(a) Why did you switch from the static model to AR(1)?

(b) Calculate the 95% confidence interval for β (the immediate effect of training).

(c) Calculate the total multiplier. Interpret it.

(d) Run the LMSC test on the AR(1) residuals. What do you conclude?

(e) Is the AR(1) model problematic given your answer to (d)?

Answer:

(a) The static model's LMSC gave $LM = 8.3 > 3.841$, rejecting H_0 — autocorrelation was detected. The static model is misspecified: the error term carries memory that should be modelled explicitly. Adding Y_{t-1} absorbs this dynamic structure.

(b) 95% CI for β : $\beta \pm 1.96 \times SE(\beta) = -0.031 \pm 1.96 \times 0.010 = -0.031 \pm 0.0196 = [-0.0506, -0.0114]$. Since 0 is outside this interval, the effect is significant at 5%. We are 95% confident that one additional training hour reduces the accident rate by between 0.011 and 0.051 incidents per 1,000 workers, immediately.

(c) Total multiplier = $\beta/(1-\gamma) = -0.031/(1 - 0.52) = -0.031/0.48 = -0.0646$. Interpretation: a permanent 1-hour increase in annual safety training reduces the long-run accident rate by approximately 0.065 incidents per 1,000 workers.

(d) $LM = 44 \times 0.021 = 0.924$. Since $0.924 < 3.841$, we fail to reject H_0 . No significant autocorrelation detected in the AR(1) residuals. A3 is plausibly satisfied. ✓

(e) No, the AR(1) model is fine here. The concern would only arise if autocorrelation were still present after adding Y_{t-1} — in that case, Y_{t-1} would be correlated with ε_t , making OLS biased. Since the LMSC test on the AR(1) residuals passes, the model is appropriately specified.

Question 5 — Full Three-Step Workflow

You have quarterly Belgian data on log(real wages) [$\ln_W$] and log(labour productivity) [$\ln_P$], $n = 80$ observations. Time plots show both series trending upward. DF test results:

Variable	DF Version	t-statistic on Y_{t-1}	Critical value (5%)	Conclusion
$\ln_W$ — level	V2 (trend)	-2.11	-3.41	?
$\ln_P$ — level	V2 (trend)	-1.87	-3.41	?
$\Delta \ln_W$ — diff	V1 (no trend)	-8.43	-2.86	?
$\Delta \ln_P$ — diff	V1 (no trend)	-7.91	-2.86	?

After differencing and fitting DL(q) with $q_{\max} = 2$, the optimal model is DL(1):

$$\Delta \ln_t = 0.002 + 0.61 \Delta P_t + 0.22 \Delta P_{t-1}$$

LMSC auxiliary regression: $n_{\text{aux}} = 78$, $R^2_{\text{aux}} = 0.009$

(a) Fill in the 'Conclusion' column of the DF table above.

(b) Why do you use first differences rather than levels?

(c) Interpret $\beta_{\text{PP}} = 0.61$ in context.

(d) Compute the total multiplier. Interpret it.

(e) Conduct the LMSC test. Conclude.

Answer:

(a) $\ln_W$ level: $-2.11 > -3.41 \rightarrow$ Fail to reject $H_0 \rightarrow$ unit root. $\ln_P$ level: $-1.87 > -3.41 \rightarrow$ Fail to reject $H_0 \rightarrow$ unit root. $\Delta \ln_W$: $-8.43 < -2.86 \rightarrow$ Reject $H_0 \rightarrow$ stationary ✓. $\Delta \ln_P$: $-7.91 < -2.86 \rightarrow$ Reject $H_0 \rightarrow$ stationary ✓.

(b) Both series have unit roots (stochastic trends). Using them in levels would risk spurious regression — t-statistics and R^2 would not have their standard interpretations. First differencing renders both series stationary, allowing valid inference.

(c) $\beta_{\Delta \Delta} = 0.61$: a 1 percentage point increase in labour productivity growth this quarter is associated with a 0.61 percentage point increase in real wage growth this same quarter, holding the lagged change in productivity constant.

(d) Total multiplier = $0.61 + 0.22 = 0.83$. Interpretation: a permanent 1 pp increase in quarterly productivity growth is associated with a long-run 0.83 pp increase in real wage growth per quarter.

(e) $LM = 78 \times 0.009 = 0.702$. Since $0.702 < 3.841$, we fail to reject $H_0: \rho = 0$. No significant autocorrelation detected. Assumption A3 is plausibly satisfied. The model is well-specified. ✓

ONE-PAGE SUMMARY

FOUNDATIONS	Regression: $Y = \alpha + \beta X + \varepsilon$. OLS finds best-fitting β . R^2 measures fit (0-1). t-statistics test if $\beta \neq 0$. p-value $< 0.05 \rightarrow$ significant. 95% CI = $\beta \pm 1.96 \times SE$. If 0 not in CI $\rightarrow$ significant. Five classical assumptions must hold.
RECOGNISE	Time series: order matters. Define lags $X_{\{t-k\}}$ and differences ΔX_t . Watch sample size ($n > 100$ ideal). Use real monetary values.
STABILISE	Time plot + Dickey-Fuller on every variable. Deterministic trend ($\lambda \neq 0$, $ \rho < 1$)? Add t as regressor. Stochastic trend / unit root ($ \rho = 1$)? Take first differences. DF uses special critical values (-2.86 or -3.41), NOT ± 1.96 . ρ SIZES: $\rho = 0$ (none) $0 < \rho \leq 0.3$ (weak) $0.3 < \rho \leq 0.6$ (moderate) $0.6 < \rho < 1$ (strong) $\rho \rightarrow 1$ (near unit root).
MODEL	Fit DL(q) or AR(1) on stationary variables. DL(q): top-down selection of q ; total multiplier = $\sum \beta_k$. AR(1): total multiplier = $\beta / (1 - \gamma)$; requires $ \gamma < 1$. Interpret: temporary vs. maintained, log-returns vs. units.
DIAGNOSE	LMSC test: auxiliary regression with lagged residual. $LM = n \cdot R^2 \sim \chi^2(1)$. Critical value = 3.841 at 5%. Reject $H_0 \rightarrow$ autocorrelation $\rightarrow$ re-specify (prefer DL over AR). DW statistic: quick check only (NOT valid for AR models).
PITFALL GRID	DF test: Reject H_0 = STATIONARY (H_0 is unit root). LMSC: Reject H_0 = AUTOCORRELATION PRESENT (H_0 is none). These are OPPOSITE conclusions from rejecting H_0 !